

Supplemental Appendix to: Self-Control and Early Withdrawal from Retirement Accounts

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A Supplemental Appendix

The fiscal implications of superannuation accounts are considerable: in Australia, tax concessions on retirement accounts cost the Federal government over 2% of GDP annually, rivaling expenditure on the means-tested public pension.¹

A.1 Tax concessions dissipated by the early release program

To gauge the amount of past tax concessions effectively dissipated by the early release program, we compute, for each withdrawer, an estimate of the concessional taxes embodied in their super balance prior to withdrawal. Aggregated using 2020-21 longitudinal survey weights, this gives a back-of-the-envelope aggregate estimate of past concessions embodied in super balances among withdrawers. We apply this to the total withdrawals—\$36.4 bn ([Australian Prudential Regulation Authority, 2021](#))—to find the total tax concessions dissipated by the program.

We calculate an estimate of past tax concessions contained in early withdrawals using HILDA data.² The actual tax concession embodied in a dollar withdrawn is a complicated object, depending on the mix of concessional and non-concessional contributions into an individual's super account, their accumulated returns, and their marginal tax rate at the time of these events. To arrive at an exact number we would require much more information than we have, and so we instead approximate it at the individual level using a calculation that focuses on the main concession, at the point of contribution.

For each individual, we first estimate the contribution-stage tax concession rate associated with concessional superannuation contributions. Let y_{it} denote taxable income

¹This aggregate is about \$60bn, the sum of items C1 to C4 in Table 1.2 of [Australian Treasury \(2025\)](#).

²See Section 3 of the main paper for a description of the data. The calculations here use the full working-age sample rather than the estimation sample in the paper.

excluding concessional employer superannuation contributions (total wages and salaries in HILDA), let c_{it} denote concessional contributions (calculated as 9.5% of total wages and salaries to approximate the compulsory employer contributions i.e. Super Guarantee), and let $T_t(\cdot)$ denote the personal income tax schedule in year t .³ The tax concession on concessional contributions in year t is approximated as

$$K_{it} = T_t(y_{it} + c_{it}) - (T_t(y_{it}) + \tau^S c_{it})$$

where $\tau^S = 15\%$ is the contributions tax rate inside superannuation. We then define the individual's average contribution-stage concession rate over the pre-pandemic period years T 2014 to 2019 as

$$\kappa_i = \frac{\sum_{t=2014}^{2019} K_{it}}{\sum_{t=2014}^{2019} (1 - \tau^S) c_{it}}$$

This gives us an estimate of the proportion of contributions to an account that are due to tax concessions. If this rate is (a) stable over a person's life and (b) equivalent to the proportion of tax concessions in retained earnings, then it also approximates the proportion of the balance.

We aggregate these proportions among withdrawers using 2020 survey weights to find the average proportion of withdrawer super balances that are due to tax concessions

$$\chi = \sum_{i \in \mathcal{N}} w_i \kappa_i$$

where w_i are 2020-21 longitudinal survey weights and $\mathcal{N}$ is the subset of the sample that withdrew under the Early Release program.⁴ χ is the proportion of total withdrawals that are past tax concessions, and we estimate its value to be $\hat{\chi} = 15.9\%$. The total value of tax concessions dissipated by the early release program is therefore found by

$$\mathcal{T} = \hat{\chi} \times \$36.4\text{bn} = \$5.8\text{bn}$$

This estimate misses the concessions on returns made inside super since contribution, although these will be roughly proportional to the contributions concession in the short run, and in the longer run will be greater. The greater weakness in the approach is the assumption of a stable concession share of contributions. In reality this will grow, on average, over a person's lifecycle as their income climbs through the progressive tax brackets. This lifecycle effect will bias our κ_i estimate up, over-estimating the true concession share of super balances. Our approach also misses non-concessional contributions (made from

³Note that the Super Guarantee rate has changed over time. It was 9.5% for the span of years relevant to this analysis, but has since increased and now stands at 12%.

⁴Ideally the weights would be each person's withdrawal as a proportion of total withdrawn. Since the modal withdrawal was the maximum, using survey weights approximates this ideal well.

after-tax income), for which no contribution-stage concession applies. These contributions are uncommon in the working-age population and concentrated among higher-income individuals near retirement (groups that are underrepresented among withdrawers in our data) so we expect this omission to introduce only a small upward bias in our estimate.

We interpret $\mathcal{T}$ as the amount of contribution-stage tax concessions associated with balances that were removed from the superannuation system before preservation age. This is not necessarily equivalent to complete dissipation of the retirement-saving purpose of those concessions. To the extent withdrawn funds were re-saved outside superannuation or used to strengthen household balance sheets, $\mathcal{T}$ overstates the policy-purpose loss. To the extent withdrawn funds were consumed, the concession value was more clearly dissipated. [Hamilton et al. \(2024\)](#) document that approximately 40% of withdrawn balances were consumed within two months, suggesting that a substantial, but not complete, share of the concession value was dissipated in this stricter sense.

A.2 Brief self-control Scale

Figure 1 shows the distribution of responses to the 13 items included in the Brief Self-Control Scale. Respondents are asked to rate how well each statement describes them, with responses ranging from 1 (“not at all”) to 5 (“very well”).

We use Principal Components Analysis (PCA) to reduce variation in the 13 BSCS items to one dimension. Table A.2 shows the estimated factor loadings for each of the 13 items. Overall, we see that the factor loadings go in the directions that we would expect based on the wording of each item. Further, the estimated factor loadings, while relatively broad-based, are largest for items related to temptation and impulsive behavior.

Table 1: PCA Factor Loadings

	Question	Loading
a	I am good at resisting temptation	-0.2772
b	I have a hard time breaking bad habits	0.2916
c	I am lazy	0.2702
d	I say inappropriate things	0.2674
e	I do certain things that are bad for me, if they are fun	0.3185
f	I refuse things that are bad for me	-0.2331
g	I wish I had more self-discipline	0.3185
h	People would say I have iron self-discipline	-0.2100
i	Pleasure and fun sometimes keep me from getting work done	0.2656
j	I have trouble concentrating	0.2930
k	I can work effectively towards long-term goals	-0.2143
l	Sometimes I cannot stop myself from doing something, even if I know it is wrong	0.3247
m	I often act without thinking through all the alternatives	0.2907

Response rates for the Brief Self-Control Scale are high. The 13 questions are included in HILDA’s self-completion questionnaire (SCQ), which is a 20 page survey consisting of a variety of questions that are difficult to administer quickly in a personal interview. Conditional on meeting our other sample requirements, 94.4% of individuals complete the SCQ, and 97.4% of SCQ respondents answer all 13 questions of the BSCS.

Of course, the Brief self-control Scale is not the only way to measure self-control issues. In general, there are two distinct approaches to measuring self-control, summarized by [Cobb-Clark et al. \(2022\)](#). The first relies upon responses to validated batteries of questions, following the canonical approach for measuring personality traits in the literature on personality psychology and economics (e.g. [Almlund et al., 2011](#); [Borghans et al., 2008](#); [Heckman et al., 2021](#)). The second approach is based on experimental economics, often measured on university students, which structurally estimates an individual’s level of self-control based on their present-bias parameter β when estimating a $\beta - \delta$ model based on incentivized tasks (e.g. [Andreoni and Sprenger, 2012](#); [Andreoni et al., 2015](#); [Augenblick and Rabin, 2019](#); [Augenblick et al., 2015](#)). In the present paper, we adopt the former approach using survey-based measurement. One benefit of this approach is that it can be embedded in large-scale household panel surveys that are nationally-representative and record a range of important economic outcomes. Both the Australian HILDA and German SOEP have recently incorporated such survey-based measurement of self-control into their large-scale panel surveys using the Brief self-control Scale.

A.3 Summary Statistics

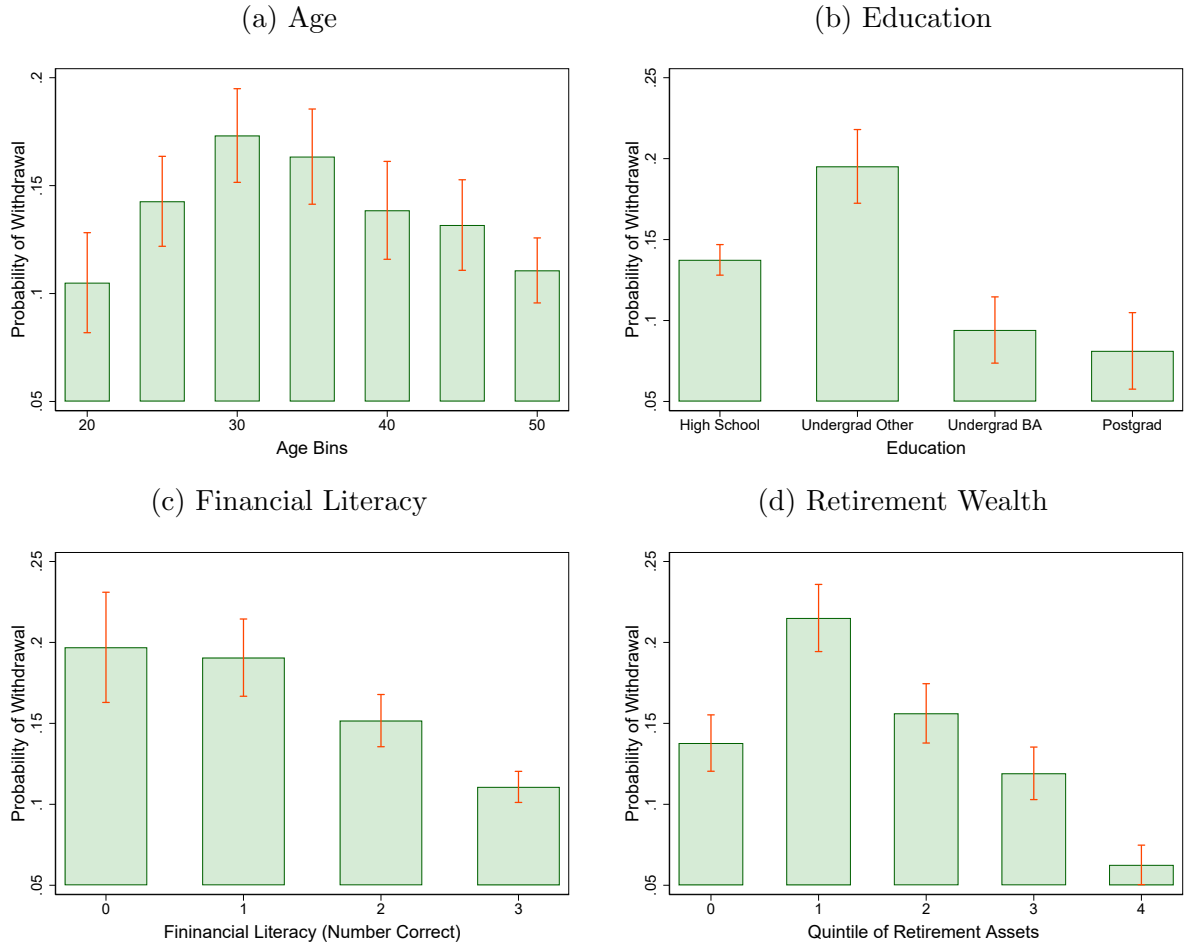
Table 2 reports averages for all of our control variables in aggregate, and compares withdrawing and non-withdrawing respondents.

Figure 2 shows the probability of early withdrawal conditional on various observable characteristics not shown in the main text. The probability of early withdrawal is highest for individuals in their thirties, which likely owes to the fact that these individuals have had time to accumulate wealth in their superannuation account, but still are early in their life-cycle and therefore may be more exposed to other shocks. Turning towards education, we see that the probability of early withdrawal is lower for those who have completed a bachelor’s or postgraduate degree. The highest probability of early withdrawal is for those classified as “Undergrad Other,” which reflects a number of undergraduate degrees including diplomas, certificates, and associate degrees, but not bachelor degrees.

Table 2: Summary Statistics

	No Withdrawal 6,538 (86.2%)	Early Withdrawal 1,047 (13.8%)	Total 7,585 (100.0%)
Unemployed			
0	87.9%	74.5%	86.0%
1	12.1%	25.5%	14.0%
incomeChangeCovidTrunc			
Not Decreased	85.1%	65.4%	82.4%
Income Loss from Covid	14.9%	34.6%	17.6%
Financial Literacy			
0	44.1%	56.5%	45.8%
1	55.9%	43.5%	54.2%
Self-Control Issues	0.390	0.423	0.394
bscs (standardized)	-0.003	0.213	0.027
Scores for component 1 (standardized)	-0.001	0.204	0.027
Age	39.210	38.305	39.085
educBins			
High School	67.5%	67.1%	67.4%
Postgraduate	7.3%	4.0%	6.8%
Undergraduate Bachelor	10.9%	7.1%	10.4%
Undergraduate Other	14.4%	21.8%	15.4%
male	0.450	0.489	0.455
Income	77,625.840	58,345.173	75,021.767
netLiquidWealth	79,730.663	29,160.334	72,750.158
netIlliquidWealth	608,484.840	309,511.091	567,215.820

Figure 2: Probability of Early Withdrawal



Note: Each figure shows the probability of early withdrawal based on a different observable characteristic. Retirement wealth is defined as the wealth held in one's superannuation account.

Turning towards financial literacy, we see that the probability of early withdrawal is declining with the number of correct answers to the “big three” financial literacy questionnaire. Finally, turning towards wealth held in superannuation accounts, we see that the probability of early withdrawal is highest for those in the low-middle part of the distribution. Individuals in the bottom quintile have very little money to withdraw. Individuals in the top quintile are relatively wealthy and may have other forms of wealth that they can draw on before turning to retirement assets. Motivated by these results, we include all of these variables as additional explanatory factors in our empirical specification.

A.4 Empirical Analysis

Table 3 reports the marginal effects for the full set of covariates included in our empirical specifications, including those omitted from the paper for the sake of expositional clarity.

Education initially appears to be an important predictor of withdrawal, although we find that most of this effect disappears once we control for wealth in specification (4). Further, although age appears strongly correlated with withdrawal in Figure 2, we find it is not an important predictor of withdrawal once we control for other factors.

We investigate the importance of the ‘Big Five’ personality traits, which have been shown to be an important predictor of labor market outcomes (see e.g. [Almlund et al., 2011](#); [Borghans et al., 2008](#); [Heckman et al., 2021](#); [Todd and Zhang, 2020](#)).⁵ Overall, we find that most of these traits are unimportant when it comes to predicting early withdrawals. Of the big five traits, only emotional stability has a significant relationship, with greater emotional stability being correlated with reduced withdrawals. That said, none of the other traits have any significant relationship with withdrawal.

In specification (4), we also control for the presence of a mortgage and the size of mortgage payments, given the possibility that early withdrawal might be more likely for mortgagors. We find no evidence of such an effect conditional on our other controls.

Table 3: Marginal Effects

	(1)	(2)	(3)	(4)
Self-Control Issues	0.16*** (0.036)	0.13*** (0.034)	0.11*** (0.036)	0.086** (0.034)
Log Income	-0.035*** (0.007)	-0.022*** (0.006)	-0.016*** (0.006)	-0.0097 (0.006)
Postgraduate	-0.055*** (0.020)	-0.052*** (0.018)	-0.041** (0.019)	-0.034 (0.022)

⁵While the use of the Big Five personality traits in explaining economic outcomes is now well-established among economists, there is much less evidence on the role of self-control, perhaps because self-control has only recently been incorporated into large-scale household surveys.

Undergraduate Bachelor	−0.051*** (0.017)	−0.046*** (0.017)	−0.038** (0.017)	−0.031* (0.017)
Undergraduate Other	0.035* (0.019)	0.033* (0.019)	0.031* (0.018)	0.019 (0.017)
Children: 1	0.070*** (0.024)	0.078*** (0.025)	0.069*** (0.024)	0.060** (0.023)
Children: 2	0.067*** (0.020)	0.067*** (0.020)	0.064*** (0.020)	0.055*** (0.019)
Children: 3+	0.11*** (0.022)	0.11*** (0.021)	0.10*** (0.020)	0.085*** (0.019)
agebins=30	0.016 (0.027)	0.015 (0.027)	0.019 (0.025)	−0.00085 (0.025)
agebins=40	−0.021 (0.024)	−0.020 (0.025)	−0.0074 (0.023)	−0.012 (0.023)
agebins=50	−0.043* (0.023)	−0.042* (0.024)	−0.025 (0.023)	−0.017 (0.023)
male	0.033** (0.013)	0.027** (0.013)	0.033** (0.013)	0.033*** (0.013)
hasPartner	−0.028* (0.016)	−0.021 (0.016)	−0.017 (0.015)	−0.0034 (0.015)
incomeMissing	−0.41*** (0.076)	−0.27*** (0.070)	−0.21*** (0.064)	−0.15** (0.064)
Income Loss from Covid		0.19*** (0.023)	0.18*** (0.021)	0.19*** (0.021)
Unemployed		0.068*** (0.016)	0.066*** (0.015)	0.058*** (0.016)
Financial Literacy			−0.042*** (0.013)	−0.028** (0.012)
Planning Horizon: Few Months			−0.031* (0.018)	−0.012 (0.017)
Planning Horizon: 1-4 Years			−0.058*** (0.018)	−0.023 (0.016)
Planning Horizon: 5+ Years			−0.065*** (0.020)	−0.023 (0.019)
Big Five: Extroversion			0.016 (0.014)	0.016 (0.014)
Big Five: Agreeableness			0.0082 (0.018)	0.014 (0.018)
Big Five: Conscientiousness			0.014 (0.017)	0.021 (0.017)
Big Five: Emotional stability			−0.033** (0.017)	−0.033** (0.017)
Big Five: Openness			−0.0035 (0.015)	−0.015 (0.015)
Liquid Assets: 2nd Quartile				−0.079*** (0.017)
Liquid Assets: 3rd Quartile				−0.12*** (0.017)
Liquid Assets: Top Quartile				−0.11*** (0.022)
Illiquid Assets: 2nd Quartile				0.017 (0.018)
Illiquid Assets: 3rd Quartile				−0.032* (0.020)

Illiquid Assets: Top Quartile				−0.049** (0.020)
Super Assets: 2nd Quartile				0.039** (0.018)
Super Assets: 3rd Quartile				0.023 (0.019)
Super Assets: Top Quartile				−0.013 (0.019)
mortgagePositive				0.12 (0.090)
logMortgagePayment				−0.018 (0.012)
Observations	7214	7214	7214	7214
Demographics	<i>Yes</i>	<i>Yes</i>	<i>Yes</i>	<i>Yes</i>
Adverse Shocks		<i>Yes</i>	<i>Yes</i>	<i>Yes</i>
Psych Controls			<i>Yes</i>	<i>Yes</i>
Wealth Controls				<i>Yes</i>
Standard errors in parentheses				
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$				

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Figure 1: Brief self-control scale questions and answers

